Service Bulletin

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Closed Crankcase Ventilation System Engine Lubricating Oil Carryover and Associated Turbocharger Lubricating Oil Leak	

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General Information

This service bulletin applies to all engines equipped with closed crankcase ventilation system.

Closed Crankcase Ventilation System Overview

Crankcase vapors, or blowby gases, are gases that escape past the piston rings during engine cycling. These gases accumulate in the crankcase, and in an open system, vent to the atmosphere. Closed crankcase ventilation system incorporates a filter and pressure regulator system. Blowby gases evacuate from the crankcase through a hose and travel toward a closed crankcase ventilation unit. The unit then uses a pressure regulator to control crankcase pressure and a coalescing filter to remove the oil from the crankcase vapors. The filtered gases return through the intake air side of the turbocharger, while the filtered oil returns to the oil pan via drain tubes.

In an event where the system malfunctions it can cause excessive engine lubricating oil to flow into the turbocharger compressor housing and charge air cooler. This does **not** indicate a malfunctioning turbocharger.

Symptom

- Engine lubricating oil leaks from the turbocharger compressor housing or charge air cooler plumbing.
- Turbocharger compressor lubricating oil staining, oil pooling in the bottom of the turbocharger compressor housing, or lubricating oil found in the air intake piping downstream from the turbocharger.

Root Cause:

- An under-performing or restricted closed crankcase ventilation system can pass engine lubricating oil from the oil pan or crankcase into the turbocharger compressor housing and the charge air cooler.
  - Potential Causes:
    - Obstructed closed crankcase ventilation system oil drain tube
    - Malfunctioning closed crankcase ventilation system oil drain tube check valve
    - Clogged closed crankcase ventilation system filter element

- Malfunctioning closed crankcase ventilation motor (if equipped)

Resolution

- Do **not** replace the turbocharger if the **only** symptom is engine lubricating oil found in the turbocharger housing or in the intake or exhaust piping. Engine lubricating oil leaks from the turbocharger compressor (cold side) or turbine (hot side) do not indicate a malfunction of the turbocharger.
- If excessive engine lubricating oil is found in the turbocharger or charge air cooler plumbing, inspect the closed crankcase ventilation system and drain tubes. See corresponding Service Manual. Reference Procedure 003-024 in Section 3. Clean the turbocharger, charge air cooler plumbing and / or charge air cooler.
- The turbocharger compressor wheel and closed crankcase ventilation system can be coated with engine lubricating oil. A light deposit of engine lubricating oil will cause no harm to the engine or turbocharger.
- If troubleshooting is completed and root cause of the malfunction is not identified, contact Cummins® Care at 1-800-CUMMINS™ for troubleshooting direction.

Examples

Figure 1, Turbocharger Compressor Cover Oil Swirling.

Figure 1, Turbocharger Compressor Cover Oil Swirling	
Appearance:	Oil swirl pattern on inside of turbocharger compressor cover.
Cause:	Normal Closed Crankcase Ventilation system oil carryover.

Figure 1, Turbocharger Compressor Cover Oil Swirling	
Action:	None.

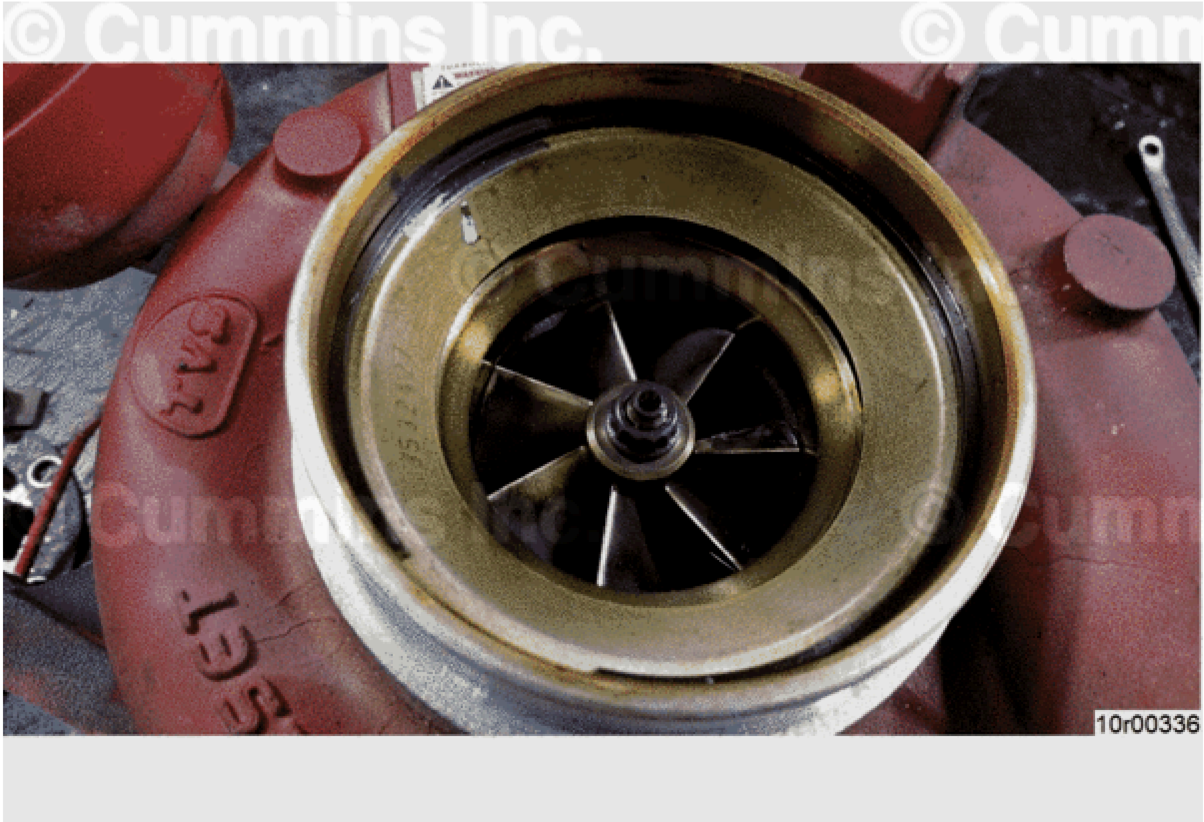

Figure 2, Turbocharger Compressor Inlet Oil Staining.

Figure 2, Turbocharger Compressor Inlet Oil Staining	
Appearance:	Oil staining on turbocharger compressor wheel and turbocharger compressor cover.
Cause:	Normal Closed Crankcase Ventilation system oil carryover.
Action:	None.

Figure 3, Turbocharger Compressor Wheel Oil Staining.

1. Turbocharger Compressor Wheel
2. Turbocharger Bearing Housing

Figure 3, Turbocharger Compressor Wheel Oil Staining	
Appearance:	Dark oil staining on turbocharger compressor wheel and turbocharger compressor cover.
Cause:	Normal Closed Crankcase Ventilation system oil carryover.
Action:	None.

Figure 4, Turbocharger Compressor Cover Oil Staining.

Figure 4, Turbocharger Compressor Cover Oil Staining	
Appearance:	Dark oil staining on inside of turbocharger compressor cover.
Cause:	Normal Closed Crankcase Ventilation system oil carryover.
Action:	None.

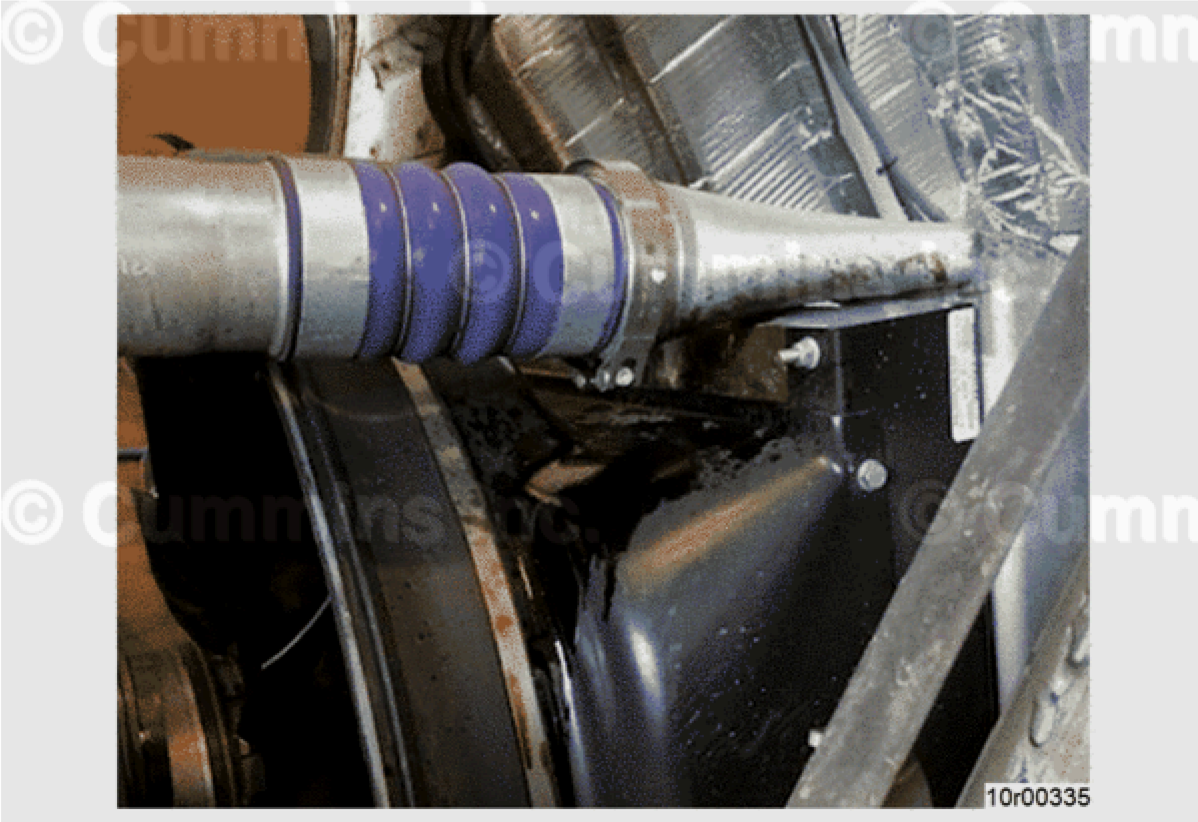

Figure 5, Charge Air Cooler Plumbing Oil Leaks.

Figure 5, Charge Air Cooler Plumbing Oil Leaks	
Appearance:	Excessive engine lubricating oil leaking from charge air cooler piping.
Cause:	Closed Crankcase Ventilation system malfunction.
Action:	Check the closed crankcase ventilation system and drain tubes. See corresponding Service Manual. Reference Procedure 003-024 in Section 3. Clean the turbocharger, charge air cooler plumbing and / or charge air cooler.

Figure 6, Turbocharger Compressor Outlet Oil Pooling.

Figure 6, Turbocharger Compressor Outlet Oil Pooling	
Appearance:	Excessive engine lubricating oil pooling at turbocharger compressor outlet
Cause:	Closed Crankcase Ventilation system malfunction.
Action:	Check the closed crankcase ventilation system and drain tubes. See corresponding Service Manual. Reference Procedure 003-024 in Section 3. Clean the turbocharger, charge air cooler plumbing and / or charge air cooler.

Document History

Date	Details
2020-10-28	Module Created
2021-7-12	Updated formatting.

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